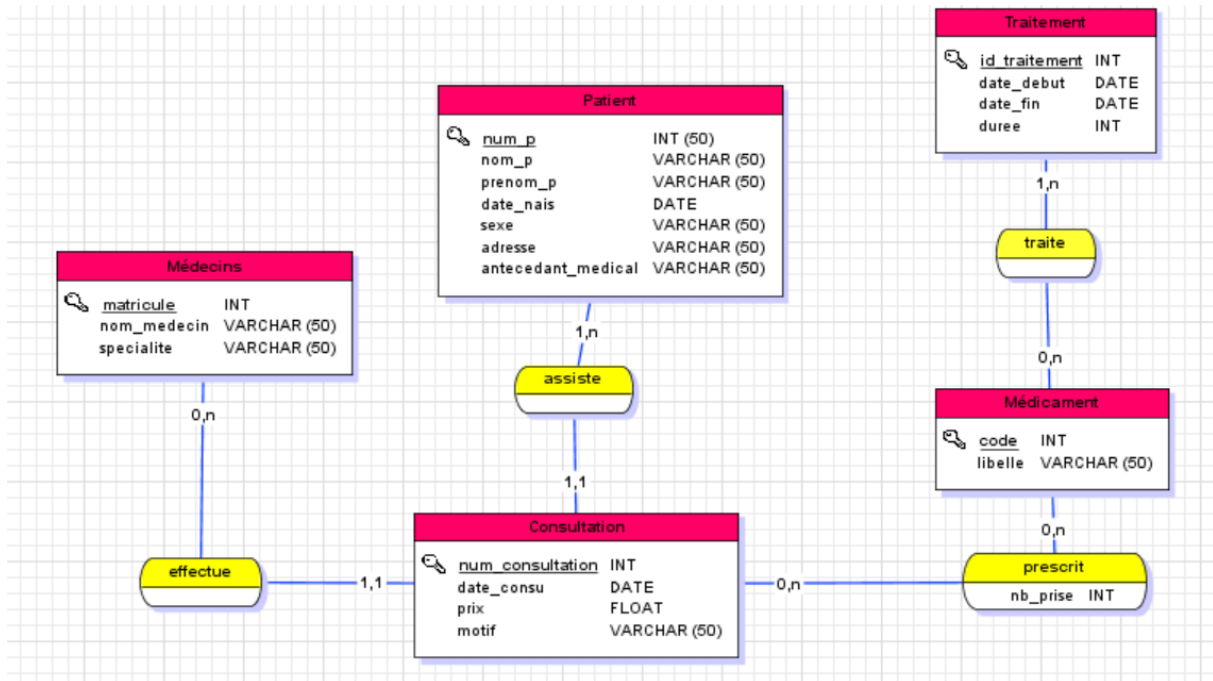


TP noté en SQL : Centre Hospitalier

ZOGLAMI HAKIM & HASSAK AREZKI

Première étape : modèle conceptuel des données



Deuxième étape : Création des tables

--MEDECINS

```
CREATE TABLE Medecins(  
    matricule Int NOT NULL ,  
    nom_medecin Varchar (50) NOT NULL ,  
    specialite Varchar (50) NOT NULL  
    ,CONSTRAINT Medecins_PK PRIMARY KEY (matricule)  
);
```

--PATIENT

```
CREATE TABLE Patient(  
    num_p Int NOT NULL ,  
    nom_p Varchar (50) NOT NULL ,  
    prenom_p Varchar (50) NOT NULL ,  
    date_nais Date NOT NULL ,  
    sexe Varchar (50) NOT NULL ,  
    adresse Varchar (50) NOT NULL,
```

```
    antecedant_medical Varchar (50) NULL
    ,CONSTRAINT Patient_PK PRIMARY KEY (num_p)
);
```

--MEDICAMENT

```
CREATE TABLE Medicament(
    code    Int NOT NULL ,
    libelle Varchar (50) NOT NULL
    ,CONSTRAINT Medicament_PK PRIMARY KEY (code)
);
```

--TRAITEMENT

```
CREATE TABLE Traitement(
    id_traitement Int NOT NULL ,
    date_debut    Date NOT NULL ,
    date_fin      Date NOT NULL
    ,CONSTRAINT Traitement_PK PRIMARY KEY (id_traitement)
);
```

--CONSULTATION

```
CREATE TABLE Consultation(
    num_consultation Int NOT NULL ,
    date_consultation Date NOT NULL ,
    prix             Float NOT NULL ,
    motif            Varchar (50) NOT NULL ,
    num_p             Int NOT NULL ,
    matricule         Int NOT NULL
    ,CONSTRAINT Consultation_PK PRIMARY KEY (num_consultation)
    ,CONSTRAINT Consultation_Patient_FK FOREIGN KEY (num_p) REFERENCES Patient(num_p)
    ,CONSTRAINT Consultation_Medecins0_FK FOREIGN KEY (matricule) REFERENCES
Medecins(matricule)
);
```

--PRESCRIT

```
CREATE TABLE prescrit(  
    code          Int NOT NULL ,  
    num_consultation Int NOT NULL ,  
    nb_prise      Int NOT NULL  
    ,CONSTRAINT prescrit_PK PRIMARY KEY (code,num_consultation)  
    ,CONSTRAINT prescrit_Medicament_FK FOREIGN KEY (code) REFERENCES Medicament(code)  
    ,CONSTRAINT prescrit_Consultation0_FK FOREIGN KEY (num_consultation) REFERENCES  
Consultation(num_consultation)  
);
```

--TRAITE

```
CREATE TABLE traite(  
    code          Int NOT NULL ,  
    id_traitement Int NOT NULL  
    ,CONSTRAINT traite_PK PRIMARY KEY (code,id_traitement)  
    ,CONSTRAINT traite_Medicament_FK FOREIGN KEY (code) REFERENCES Medicament(code)  
    ,CONSTRAINT traite_Traitement0_FK FOREIGN KEY (id_traitement) REFERENCES  
Traitement(id_traitement)  
);
```

Troisième étape : Insertion des données

--PATIENT

```
INSERT INTO Patient VALUES (1, 'Zoghلامي', 'Hakim', '05-APR-2000', 'M', 'Marseille 15',  
'Cardiomyopathie');  
  
INSERT INTO Patient VALUES (2, 'Martin', 'Sophie', '22-MAR-1985', 'F', 'Marseille 13', 'Asthme');  
  
INSERT INTO Patient VALUES (3, 'Bamcolo', 'Luqman', '01-DEC-1978', 'M', 'Marseille 15', 'Diabète');  
  
INSERT INTO Patient VALUES (4, 'Boussoumart', 'Emma', '08-JUL-1995', 'F', 'Marseille 16', 'Allergie');  
  
INSERT INTO Patient VALUES (5, 'Fasfat', 'Sophiane', '17-NOV-1982', 'M', 'Marseille 14', 'Migraine');  
  
INSERT INTO Patient VALUES (6, 'Abou', 'Meredhit', '05-APR-1999', 'F', 'Marseille 07', NULL);  
  
INSERT INTO Patient VALUES (7, 'Ali', 'Farid', '29-SEP-2000', 'M', 'Marseille 09', 'cardiomyopathie');  
  
INSERT INTO Patient VALUES (8, 'Petit', 'Claire', '15-JAN-1987', 'F', 'Marseille 10', 'Asthme');  
  
INSERT INTO Patient VALUES (9, 'Lemoine', 'Julien', '10-OCT-1992', 'M', 'Marseille 11', NULL);
```

```

INSERT INTO Patient VALUES (10, 'Garcia', 'Lucie', '22-FEB-1995', 'F', 'Lille', 'Allergie');
INSERT INTO Patient VALUES (11, 'Ben Amar', 'Yacine', '15-MAY-1995', 'M', 'Marseille 14', 'Diabète');
INSERT INTO Patient VALUES (12, 'Bensalem', 'Khadija', '22-APR-1980', 'F', 'Marseille 13',
'Hypertension');
INSERT INTO Patient VALUES (13, 'Benkacem', 'Mounir', '01-JAN-2002', 'M', 'Marseille 15',
'Cardiomyopathie');
INSERT INTO Patient VALUES (14, 'Saidi', 'Fatima', '08-MAR-1990', 'F', 'Marseille 10', NULL);
INSERT INTO Patient VALUES (15, 'Zeroual', 'Ahmed', '25-DEC-1978', 'M', 'Marseille 16',
'Cardiomyopathie');
INSERT INTO Patient VALUES (16, 'Bouchaib', 'Imane', '05-SEP-1988', 'F', 'Marseille 14', 'Migraine');
INSERT INTO Patient VALUES (17, 'Cherif', 'Nassim', '12-OCT-1997', 'M', 'Marseille 13', 'Allergie');
INSERT INTO Patient VALUES (18, 'Khelifi', 'Salma', '18-AUG-1985', 'F', 'Marseille 11', 'Asthme');
INSERT INTO Patient VALUES (19, 'Mokhtar', 'Anis', '30-JUL-1992', 'M', 'Marseille 9', NULL);
INSERT INTO Patient VALUES (20, 'Boumediene', 'Lina', '10-NOV-2000', 'F', 'Marseille 12',
'Cardiomyopathie');
SELECT * FROM Patient;

```

--MEDECIN

```

INSERT INTO Medecins VALUES (101, 'Leclerc', 'Cardiologie');
INSERT INTO Medecins VALUES (102, 'Moreau', 'Dermatologie');
INSERT INTO Medecins VALUES (103, 'Bernard', 'Pédiatrie');
INSERT INTO Medecins VALUES (104, 'Simon', 'Gastroentérologie');
INSERT INTO Medecins VALUES (105, 'Giraud', 'Neurologie');
INSERT INTO Medecins VALUES (106, 'Perrin', 'Oncologie');
INSERT INTO Medecins VALUES (107, 'Dubois', 'Psychiatrie');
SELECT * FROM Medecins;

```

--CONSULTATION

```

INSERT INTO Consultation VALUES (1001, '15-JUN-2024', 50.00, 'Suivi général', 1, 101);
INSERT INTO Consultation VALUES (1002, '20-JUN-2024', 70.00, 'Contrôle cardiaque', 2, 101);
INSERT INTO Consultation VALUES (1003, '10-JUL-2024', 40.00, 'Consultation dermatologique', 3,
102);
INSERT INTO Consultation VALUES (1004, '25-JUL-2024', 60.00, 'Consultation pédiatrique', 4, 103);
INSERT INTO Consultation VALUES (1005, '05-AUG-2024', 90.00, 'Problèmes digestifs', 6, 104);
INSERT INTO Consultation VALUES (1006, '10-AUG-2024', 80.00, 'Migraines récurrentes', 5, 105);

```

```
INSERT INTO Consultation VALUES (1007, '15-JUN-2024', 45.00, 'Suivi général', 1, 101);
INSERT INTO Consultation VALUES (1008, '25-JUL-2024', 55.00, 'Contrôle annuel', 4, 103);
INSERT INTO Consultation VALUES (1009, '01-JUL-2024', 75.00, 'Diagnostic oncologique', 8, 106);
INSERT INTO Consultation VALUES (1010, '12-JUL-2024', 65.00, 'Évaluation psychiatrique', 9, 107);
INSERT INTO Consultation VALUES (1011, '15-JUN-2024', 50.00, 'Contrôle général', 11, 101);
INSERT INTO Consultation VALUES (1012, '18-JUN-2024', 60.00, 'Évaluation dermatologique', 12, 102);
INSERT INTO Consultation VALUES (1013, '20-JUN-2024', 70.00, 'Consultation pédiatrique', 13, 103);
INSERT INTO Consultation VALUES (1014, '22-JUN-2024', 80.00, 'Suivi diabète', 15, 105);
INSERT INTO Consultation VALUES (1015, '25-JUN-2024', 55.00, 'Problèmes respiratoires', 16, 104);
INSERT INTO Consultation VALUES (1016, '28-JUN-2024', 65.00, 'Consultation cardiaque', 17, 101);
INSERT INTO Consultation VALUES (1017, '01-JUL-2024', 75.00, 'Diagnostic neurologique', 18, 106);
INSERT INTO Consultation VALUES (1018, '04-JUL-2024', 90.00, 'Migraine aiguë', 19, 107);
INSERT INTO Consultation VALUES (1019, '07-JUL-2024', 85.00, 'Consultation annuelle', 20, 103);
INSERT INTO Consultation VALUES (1020, '10-JUL-2024', 100.00, 'Problème digestif', 14, 104);
SELECT * FROM Consultation;
```

--MEDICAMENT

```
INSERT INTO Medicament VALUES (201, 'Doliprane');
INSERT INTO Medicament VALUES (202, 'Ibuprofène');
INSERT INTO Medicament VALUES (203, 'Amoxicilline');
INSERT INTO Medicament VALUES (204, 'Paracétamol');
INSERT INTO Medicament VALUES (205, 'Oméprazole');
INSERT INTO Medicament VALUES (206, 'Aspirine');
INSERT INTO Medicament VALUES (207, 'Cortisone');
INSERT INTO Medicament VALUES (208, 'Morphine');
INSERT INTO Medicament VALUES (209, 'Paracétamol');
INSERT INTO Medicament VALUES (210, 'Doliprane');
SELECT * FROM Medicament;
```

--TRAITEMENT

```
INSERT INTO Traitement VALUES (301, '16-JUN-2024', '26-JUL-2024');
INSERT INTO Traitement VALUES (302, '21-JUN-2024', '30-JUL-2024');
INSERT INTO Traitement VALUES (303, '11-JUL-2024', '01-AUG-2024');
INSERT INTO Traitement VALUES (304, '02-JUL-2024', '16-JUL-2024');
INSERT INTO Traitement VALUES (305, '06-AUG-2024', '06-SEP-2024');
INSERT INTO Traitement VALUES (306, '11-AUG-2024', '11-SEP-2024');
INSERT INTO Traitement VALUES (307, '16-AUG-2024', '20-AUG-2024');
INSERT INTO Traitement VALUES (308, '01-JUL-2024', '30-DEC-2024');
INSERT INTO Traitement VALUES (309, '15-JUL-2024', '15-DEC-2024');
INSERT INTO Traitement VALUES (310, '01-JUL-2024', '01-SEP-2024');
INSERT INTO Traitement VALUES (311, '05-JUL-2024', '05-OCT-2024');
INSERT INTO Traitement VALUES (312, '10-JUL-2024', '10-JAN-2025');
INSERT INTO Traitement VALUES (313, '15-JUL-2024', '15-NOV-2024');
INSERT INTO Traitement VALUES (314, '20-JUL-2024', '20-AUG-2024');
INSERT INTO Traitement VALUES (315, '25-JUL-2024', '25-DEC-2024');
INSERT INTO Traitement VALUES (316, '01-AUG-2024', '01-NOV-2024');
INSERT INTO Traitement VALUES (317, '05-AUG-2024', '05-OCT-2024');
INSERT INTO Traitement VALUES (318, '10-AUG-2024', '10-JAN-2025');
INSERT INTO Traitement VALUES (319, '15-AUG-2024', '15-FEB-2025');
SELECT * FROM Traitement;
```

--PRESCRIT

```
INSERT INTO prescrit VALUES (201, 1001, 3);
INSERT INTO prescrit VALUES (202, 1002, 2);
INSERT INTO prescrit VALUES (203, 1003, 1);
INSERT INTO prescrit VALUES (204, 1004, 3);
INSERT INTO prescrit VALUES (205, 1005, 2);
INSERT INTO prescrit VALUES (206, 1006, 1);
INSERT INTO prescrit VALUES (207, 1009, 2);
INSERT INTO prescrit VALUES (208, 1010, 1);
INSERT INTO prescrit VALUES (201, 1011, 2);
```

```
INSERT INTO prescrit VALUES (202, 1012, 3);
INSERT INTO prescrit VALUES (203, 1013, 1);
INSERT INTO prescrit VALUES (204, 1014, 2);
INSERT INTO prescrit VALUES (205, 1015, 1);
INSERT INTO prescrit VALUES (206, 1016, 2);
INSERT INTO prescrit VALUES (207, 1017, 3);
INSERT INTO prescrit VALUES (208, 1018, 1);
INSERT INTO prescrit VALUES (209, 1019, 2);
INSERT INTO prescrit VALUES (210, 1020, 3);
SELECT * FROM prescrit;
```

--TRAITE

```
INSERT INTO traite VALUES (201, 301);
INSERT INTO traite VALUES (202, 302);
INSERT INTO traite VALUES (203, 303);
INSERT INTO traite VALUES (204, 304);
INSERT INTO traite VALUES (205, 305);
INSERT INTO traite VALUES (206, 306);
INSERT INTO traite VALUES (207, 307);
INSERT INTO traite VALUES (208, 308);
INSERT INTO traite VALUES (208, 309);
INSERT INTO traite VALUES (202, 310);
INSERT INTO traite VALUES (203, 311);
INSERT INTO traite VALUES (204, 312);
INSERT INTO traite VALUES (205, 313);
INSERT INTO traite VALUES (206, 314);
INSERT INTO traite VALUES (207, 315);
INSERT INTO traite VALUES (208, 316);
SELECT * FROM traite;
```

Quatrième étapes : requête SQL

Question 1 -----

```
SELECT num_p , nom_p , prenom_p ,  
COUNT(*) AS nombre_consultations  
FROM Patient  
NATURAL JOIN Consultation  
GROUP BY num_p, nom_p, prenom_p ;
```

9	Lemoine	Julien	1
11	Ben Amar	Yacine	1
16	Bouchaib	Imane	1
18	Khelifi	Salma	1

NUM_P	NOM_P	PRENOM_P	NOMBRE_CONSULTATIONS
3	Bamcolo	Luqman	1
14	Saïdi	Fatima	1
6	Abou	Meredhit	1
17	Chérif	Nassim	1
13	Benkacem	Mounir	1
4	Boussoumart	Emma	2
12	Bensalem	Khadija	1

19	Mokhtar	Anis	1
1	Zoghlami	Hakim	2
2	Martin	Sophie	1
5	Fasfat	Sophiane	1
8	Petit	Claire	1
20	Boumediene	Lina	1
15	Zeroual	Ahmed	1
9	Lemoine	Julien	1

Question 2 -----

```
SELECT num_p, nom_p, prenom_p  
FROM Patient  
NATURAL JOIN Consultation  
NATURAL JOIN prescrit  
WHERE code = 201;
```

NUM_P	NOM_P	PRENOM_P
1	Zoghlami	Hakim
11	Ben Amar	Yacine

Question 3 -----

```
SELECT DISTINCT num_p, nom_p, prenom_p  
FROM Patient  
NATURAL JOIN Consultation  
NATURAL JOIN Medecins  
WHERE specialite = 'Cardiologie';
```

NUM_P	NOM_P	PRENOM_P
17	Chérif	Nassim
1	Zoghlami	Hakim
2	Martin	Sophie
11	Ben Amar	Yacine

Question 4

```
SELECT AVG(prix) AS prix_moyen_cardiologie
FROM Consultation
NATURAL JOIN Medecins
WHERE specialite = 'Cardiologie';
```

PRIX_MOYEN_CARDIOLOGIE
56

Question 5

```
SELECT specialite,
COUNT(num_consultation) AS nombre_consultations
FROM Consultation
NATURAL JOIN Medecins
GROUP BY specialite;
```

SPÉCIALITÉ	NOMBRE_CONSULTATIONS
Psychiatrie	2
Cardiologie	5
Pédiatrie	4
Dermatologie	2
Gastroentérologie	3
Neurologie	2
Oncologie	2

Question 6

```
SELECT antecedant_medical,
COUNT(*) AS nombre_patients
FROM Patient
GROUP BY antecedant_medical
ORDER BY nombre_patients DESC;
```

ANTECEDANT_MEDICAL	NOMBRE_PATIENTS
-	4
Cardiomyopathie	4
Allergie	3
Asthme	3
Migraine	2
Diabète	2
Hypertension	1
cardiomyopathie	1

Question 7 -----

```
SELECT num_p, nom_p, prenom_p
```

FROM Patient

NATURAL JOIN Consultation

NATURAL JOIN prescrit

NATURAL JOIN traite

NATURAL JOIN Traitement

WHERE MONTHS_BETWEEN(date_fin, date_debut) > 3;

NUM_P	NOM_P	PRENOM_P
4	Boussoumart	Emma
6	Abou	Meredhit
8	Petit	Claire
9	Lemoine	Julien
9	Lemoine	Julien
15	Zeroual	Ahmed
16	Bouchaib	Imane

18	Khelifi	Salma
19	Mokhtar	Anis
19	Mokhtar	Anis

Question 8-----

```
SELECT nom_medecin,
```

COUNT(num_consultation) / COUNT(DISTINCT date_cons) AS moyenne_consultations_par_jour

FROM Medecins

NATURAL JOIN Consultation

GROUP BY nom_medecin;

NOM_MEDECIN	MOYENNE_CONSULTATIONS_PAR_JOUR
Giraud	1
Simon	1
Dubois	1
Leclerc	1.6667
Moreau	1
Bernard	1.33
Perrin	2

Question 9-----

```
SELECT specialite,  
COUNT(num_consultation) AS nb_consultations  
FROM Consultation  
NATURAL JOIN Medecins  
WHERE date_cons BETWEEN TO_DATE('01-JUN-2024', 'DD-MON-YYYY') AND TO_DATE('31-AUG-  
2024', 'DD-MON-YYYY')  
GROUP BY specialite  
ORDER BY nb_consultations DESC;
```

SPÉCIALITÉ	NB_CONSULTATIONS
Cardiologie	5
Pédiatrie	4
Gastroentérologie	3
Neurologie	2
Oncologie	2
Psychiatrie	2
Dermatologie	2

Question 10-----

```
SELECT num_p AS patient_id, nom_p AS nom, prenom_p AS prenom, id_traitement, date_debut,  
date_fin  
FROM Patient  
NATURAL JOIN Consultation  
NATURAL JOIN prescrit  
NATURAL JOIN traite  
NATURAL JOIN Traitement  
WHERE date_fin >= SYSDATE  
GROUP BY num_p, nom_p, prenom_p, id_traitement, date_debut, date_fin  
HAVING MAX(nb_prise) > 2 OR COUNT( num_consultation) > 2;
```

PATIENT_ID	NOM	PRÉNOM	ID_TRAITEMENT	DATE_DEBUT	DATE_FIN
4	Boussoumart	Emma	312	10-JUIL-24	10-JANV-25
18	Khelifi	Salma	315	25-JUIL-24	25-DÉC-24

Question 11

```
SELECT  
  
SUM(CASE WHEN FLOOR(MONTHS_BETWEEN(SYSDATE, date_nais) / 12) BETWEEN 20 AND 30 THEN  
1 ELSE 0 END) AS "20-30 ans",  
  
SUM(CASE WHEN FLOOR(MONTHS_BETWEEN(SYSDATE, date_nais) / 12) BETWEEN 31 AND 40 THEN  
1 ELSE 0 END) AS "31-40 ans",  
  
SUM(CASE WHEN FLOOR(MONTHS_BETWEEN(SYSDATE, date_nais) / 12) BETWEEN 41 AND 50 THEN  
1 ELSE 0 END) AS "41-50 ans",  
  
SUM(CASE WHEN FLOOR(MONTHS_BETWEEN(SYSDATE, date_nais) / 12) > 50 THEN 1 ELSE 0 END) AS  
"50 ans et plus"  
  
FROM Patient;
```

20-30 ans	31-40 ans	41-50 ans	50 ans et plus
9	7	4	0

Question 12

```
SELECT libelle AS Nom_Medicament_Traitement,  
  
AVG(MONTHS_BETWEEN(Traitement.date_fin, Traitement.date_debut)) AS Duree_Moyenne  
  
FROM Traitement  
  
NATURAL JOIN traite  
  
NATURAL JOIN prescrit  
  
NATURAL JOIN Consultation  
  
NATURAL JOIN Patient  
  
NATURAL JOIN Medicament  
  
WHERE Patient.sexe = 'M'  
  
AND FLOOR(MONTHS_BETWEEN(SYSDATE, Patient.date_nais) / 12) < 25  
  
AND Patient.antecedant_medical = 'Cardiomyopathie'  
  
AND Traitement.date_fin IS NOT NULL  
  
GROUP BY libelle;
```

NOM_MEDICAMENT_TRAITEMENT	DUREE_MOYENNE
Doliprane	1.32258064516129032258064516129032258065
Amoxicilline	1.83870967741935483870967741935483870968